```python
def total_prime_factor_count(n):
    count = 0
    while n % 2 == 0:
        count += 1
        n //= 2
    p = 3
    while p * p <= n:
        while n % p == 0:
            count += 1
            n //= p
        p += 2
    if n > 1:
        count += 1
    return count
n = int(input())
print(total_prime_factor_count(n))
```

12

**OUTPUT**

```
3
```

12

```python
import math

n = int(input("Enter a number: "))

count = 0
for i in range(1, int(math.isqrt(n)) + 1):
    if n % i == 0:
        count += 1 if i * i == n else 2

print(count)
```

OUTPUT

```
Enter a number: 6
```

```python
def is_perfect_number(n):
    if n <= 1:
        return False

    total = 0
    for i in range(1, n):
        if n % i == 0:
            total += i

    return total == n

num = int(input("Enter a number: "))

if is_perfect_number(num):
    print(f"{num} is a perfect number.")
else:
    print(f"{num} is not a perfect number.")
```

28

**OUTPUT**

```
Enter a number: 28 is a perfect number.
```

```python
def digit_multiplicative_persistence(n):
    count = 0
    while n >= 10:
        product = 1
        for ch in str(n):
            product *= int(ch)
        n = product
        count += 1
    return count

num = int(input("Enter a number: "))
print("Persistence:", digit_multiplicative_persistence(num))
```

715

OUTPUT

```
Enter a number: Persistence: 3
```

```python
n = input().strip()  # e.g., 89781

result = []
for ch in n:
    d = int(ch)
    result.append(str((d + 1) % 10))  # next digit, wrap 9 -> 0

print("".join(result))
```

89781

**OUTPUT**

```
90892
```

```python
def reverse_odd_digits(n):
    s = str(n)
    odds = [ch for ch in s if int(ch) % 2 != 0]
    odds.reverse()
    res = []
    idx = 0
    for ch in s:
        if int(ch) % 2 != 0:
            res.append(odds[idx])
            idx += 1
        else:
            res.append(ch)
    return int("".join(res))
```

816253

OUTPUT

```python
def encode(s: str) -> str:
    if not s:
        return ""

    result = []
    count = 1

    for i in range(1, len(s)):
        if s[i] == s[i - 1]:
            count += 1
        else:
            result.append(f"{count}{s[i-1]}")
            count = 1

    result.append(f"{count}{s[-1]}")
    return "".join(result)

# Example
inp = "55553322255"
print("Input :", inp)
print("Encoded:", encode(inp))
```

55553322255

OUTPUT

```
Input : 55553322255
Encoded: 45233225
```

```python
s = input().strip()
digits = list(s)

even_positions = [i for i, ch in enumerate(digits) if int(ch) % 2 == 0]
even_digits = [digits[i] for i in even_positions][::-1]

for pos, d in zip(even_positions, even_digits):
    digits[pos] = d

print(''.join(digits))
```

325649

**OUTPUT**

```
345629
```

```python
def is_step_number(n: int) -> bool:
    s = str(n)
    if len(s) <= 1:
        return True  # single digit is a step number

    for i in range(1, len(s)):
        if abs(int(s[i]) - int(s[i-1])) != 1:
            return False
    return True

num = int(input("Enter a number: "))
print("YES" if is_step_number(num) else "NO")
```

54567

OUTPUT

```
Enter a number: YES
```

```python
arr = list(map(int, input("Enter elements separated by space: ").split()))

odd_count = 0
even_count = 0

odd_numbers = []
even_numbers = []

for x in arr:
    if x % 2 != 0:
        odd_numbers.append(x)
        odd_count += 1
    else:
        even_numbers.append(x)
        even_count += 1

print("Odd sequence:", *odd_numbers, "Count:", odd_count)
print("Even sequence:", *even_numbers, "Count:", even_count)
```

51336843

OUTPUT

```
Enter elements separated by space: Odd sequence: 51336843 Count: 1
Even sequence: Count: 0
```

```python
n = input().strip()  # e.g., 658134

odd = []
even = []

for ch in n:
    d = int(ch)
    if d % 2 != 0:
        odd.append(ch)
    else:
        even.append(ch)

print(''.join(odd + even))
```

658134

OUTPUT

```
513684
```

```python
n = int(input())
x = n + 1

while True:
    is_prime = True
    for i in range(2, x):
        if x % i == 0:
            is_prime = False
            break
    if is_prime:
        print(x)
        break
    x += 1
```

16

OUTPUT

```
17
```

```python
num = input("Enter a number: ")

while len(num) > 1:
    total = 0
    for digit in num:
        total += int(digit)
    num = str(total)

print("Output:", num)
```

89789

OUTPUT

```
Enter a number: Output: 5
```

```python
num = input("Enter a number: ")
total = 0
for digit in num:
    total += int(digit)
print("Sum of digits:", total)
```

89789

OUTPUT

```
Enter a number: Sum of digits: 41
```